Name: Shevchenko Mariia

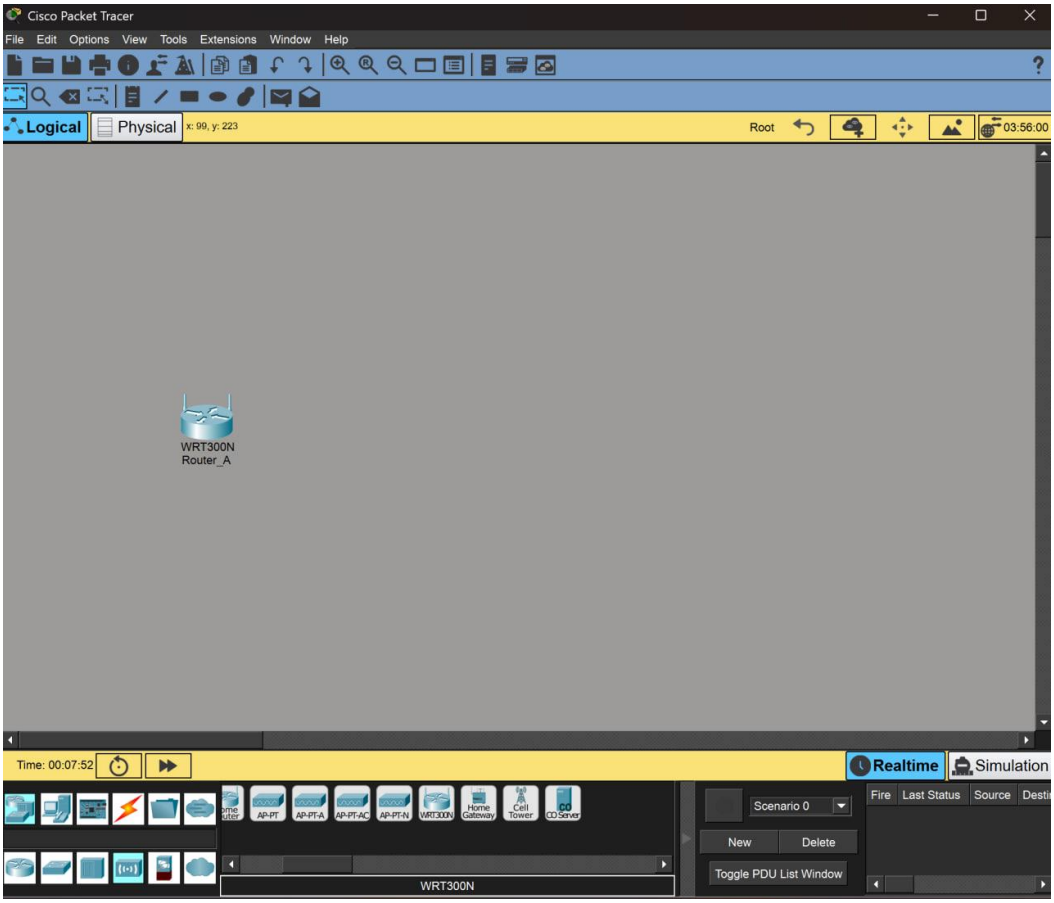
Student ID: 64594

Course: Network Security

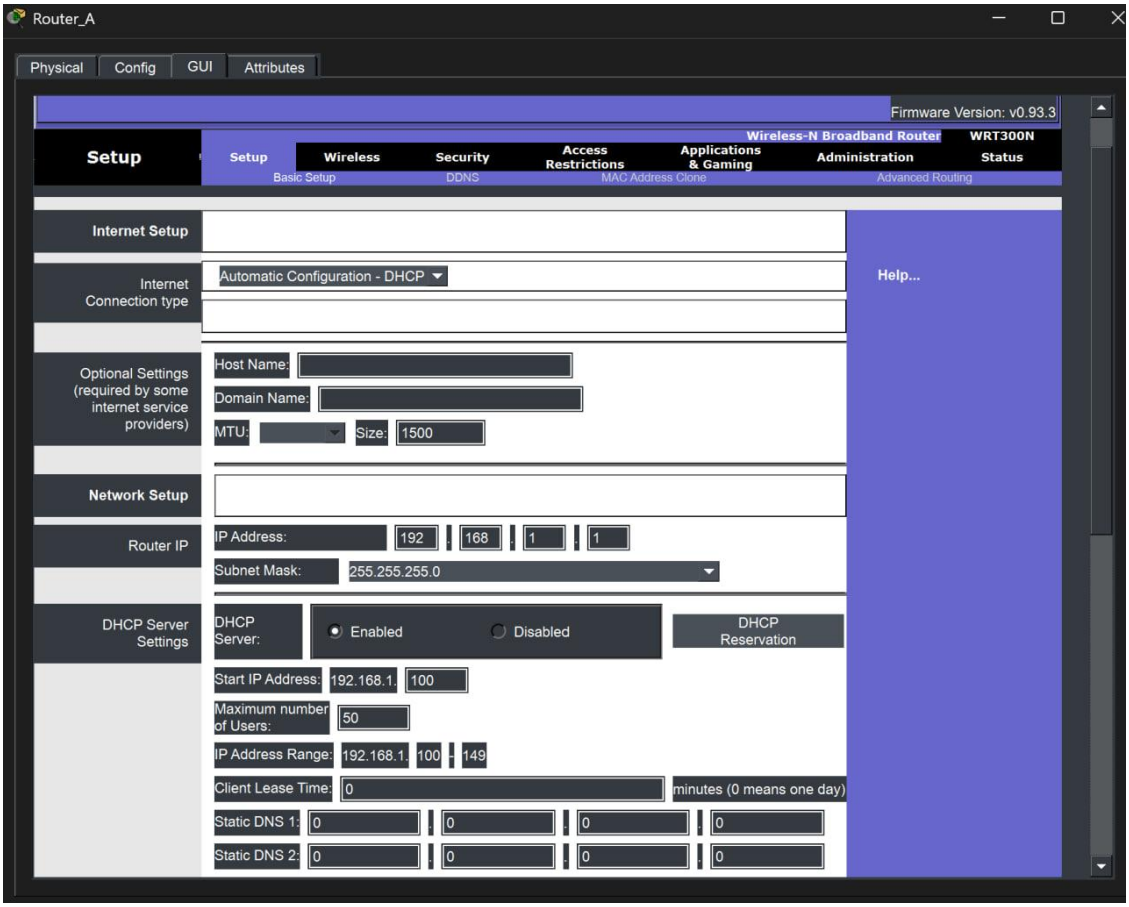
Date: 06.06.2026

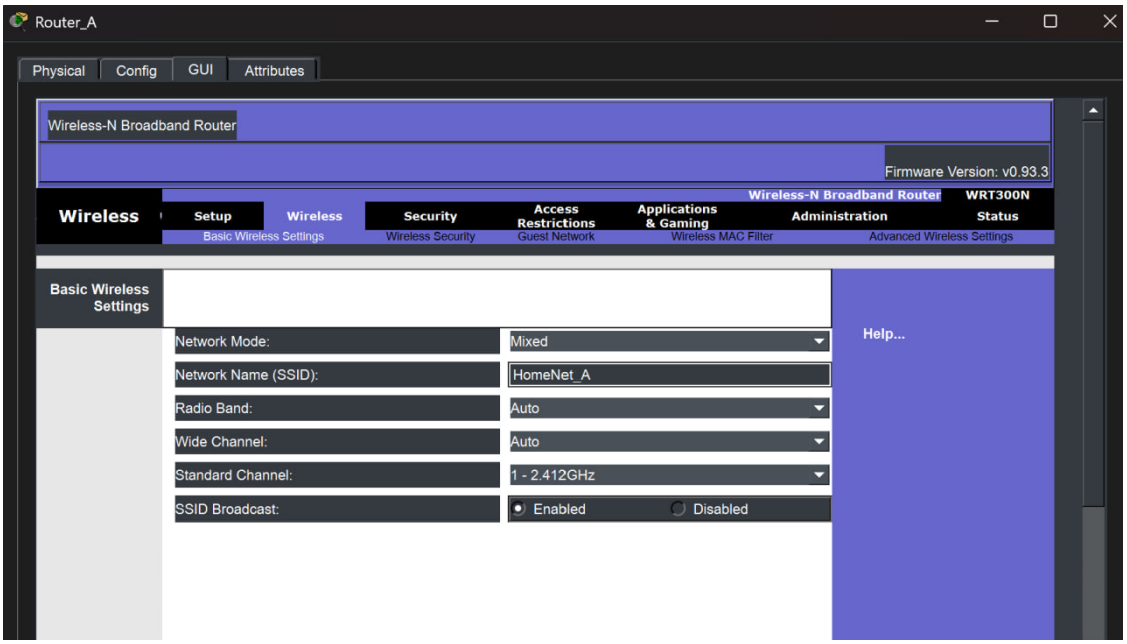
Introduction:

This laboratory assignment focuses on the practical configuration and security assessment of a wireless local area network (WLAN) using Cisco Packet Tracer. The main objective of the exercise is to build and maintain two entirely separate, independent home wireless networks. Each network uses its own unique Service Set Identifier (SSID), a distinct IP address range, and a dedicated Dynamic Host Configuration Protocol (DHCP) pool. By keeping these two environments separate, the lab demonstrates how wireless routers manage client devices within their own wireless coverage areas without interfering with each other.

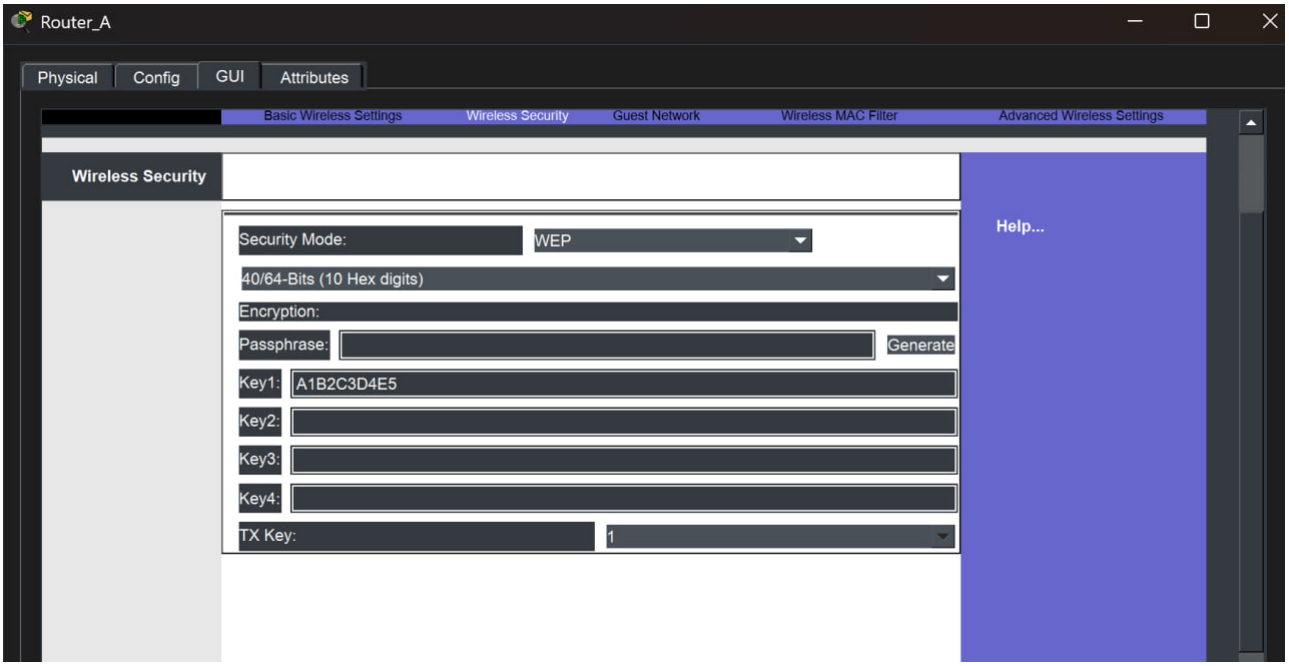


Description: This screenshot displays the initial setup of the Cisco Packet Tracer workspace. A single wireless home routing device has been added and renamed to establish the base infrastructure for Network A.

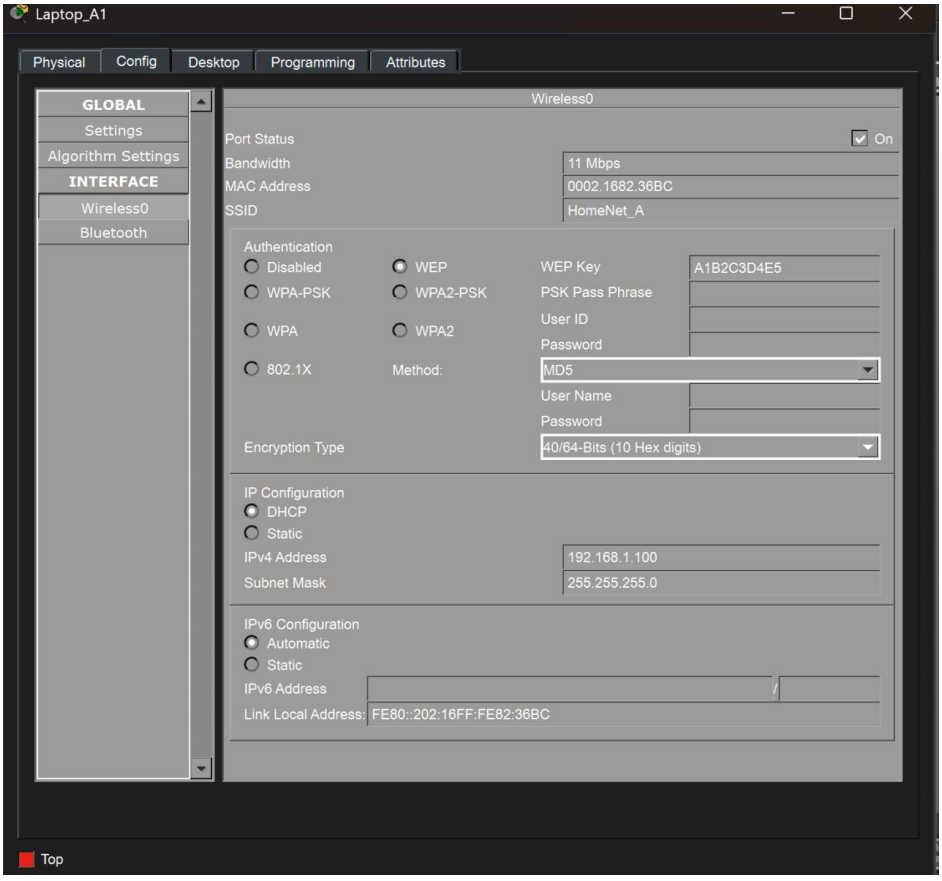


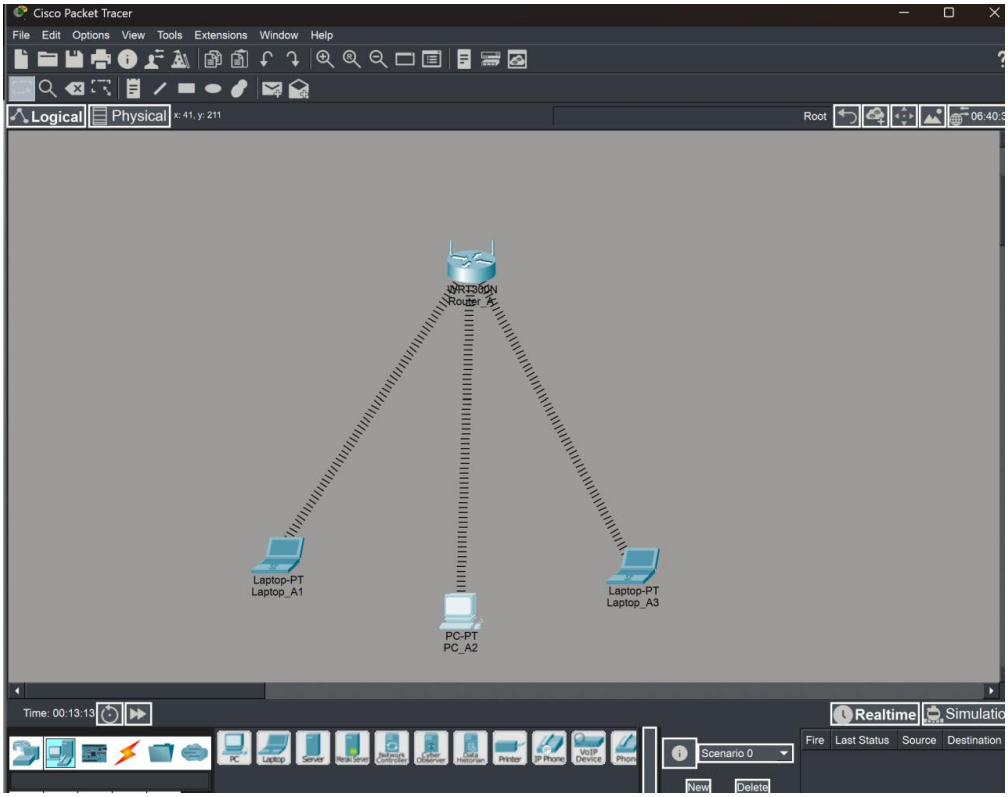


Description: This screenshot shows the basic configuration inside GUI of Router_A. The local area network IP address has been configured to 192.168.1.1 with a standard /24 subnet mask (255.255.255.0), and the local DHCP server is active. Under the wireless settings, the network name or Service Set Identifier (SSID) has been changed to HomeNet_A.

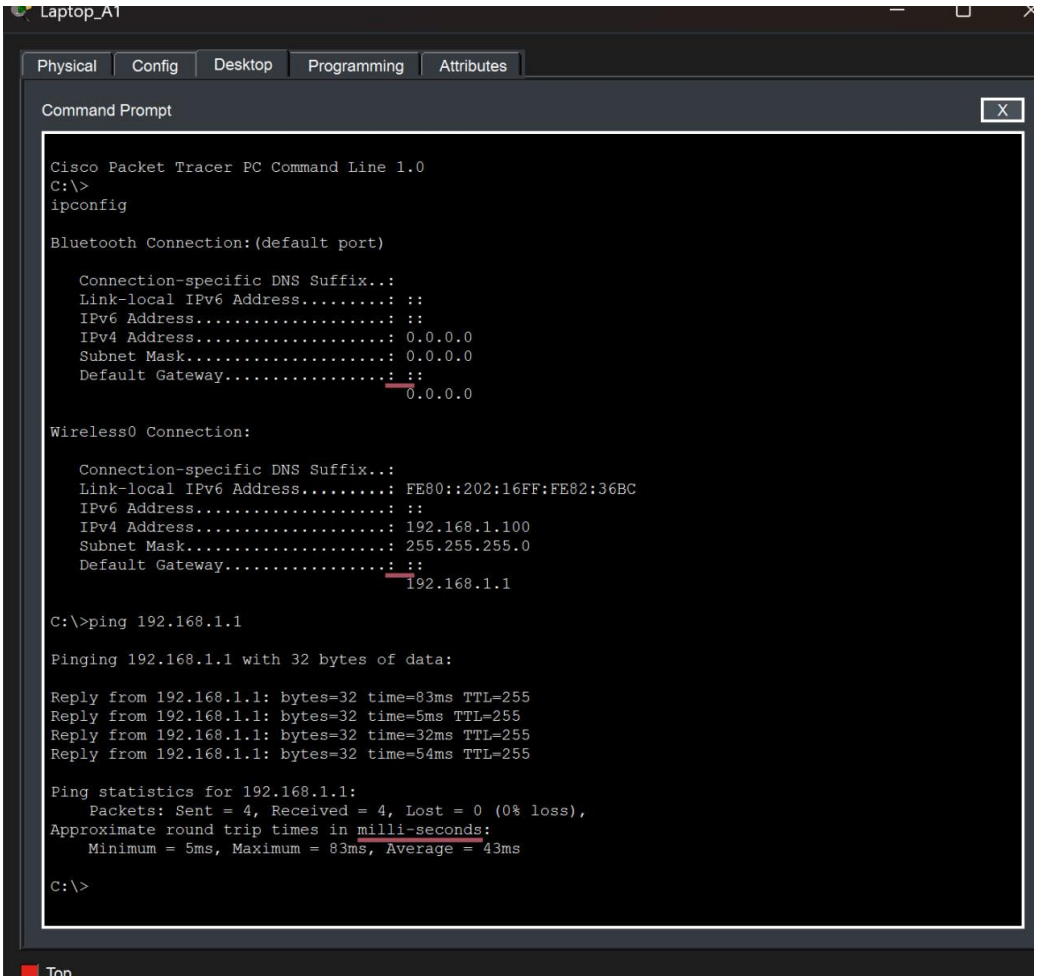


Description: This screenshot shows the security configuration on Router_A. The security mode has been changed from disabled to WEP. The encryption key has been set using Key 1 with the specific 10-character hexadecimal password A1B2C3D4E5 under open system authentication.

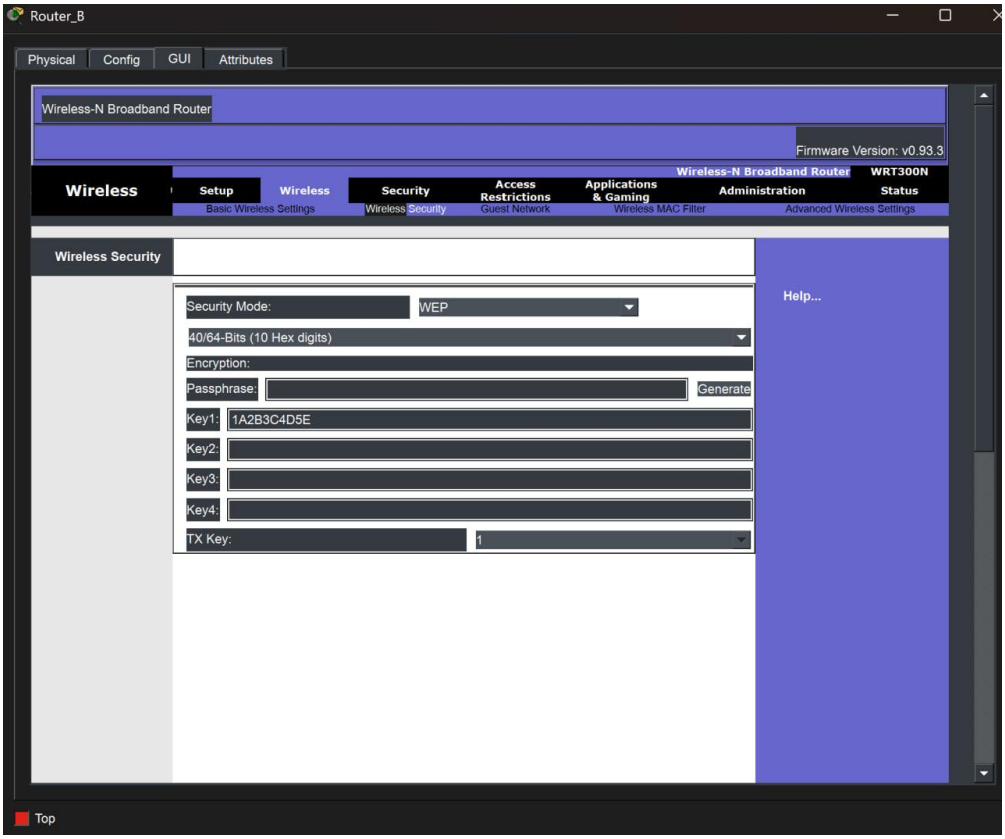


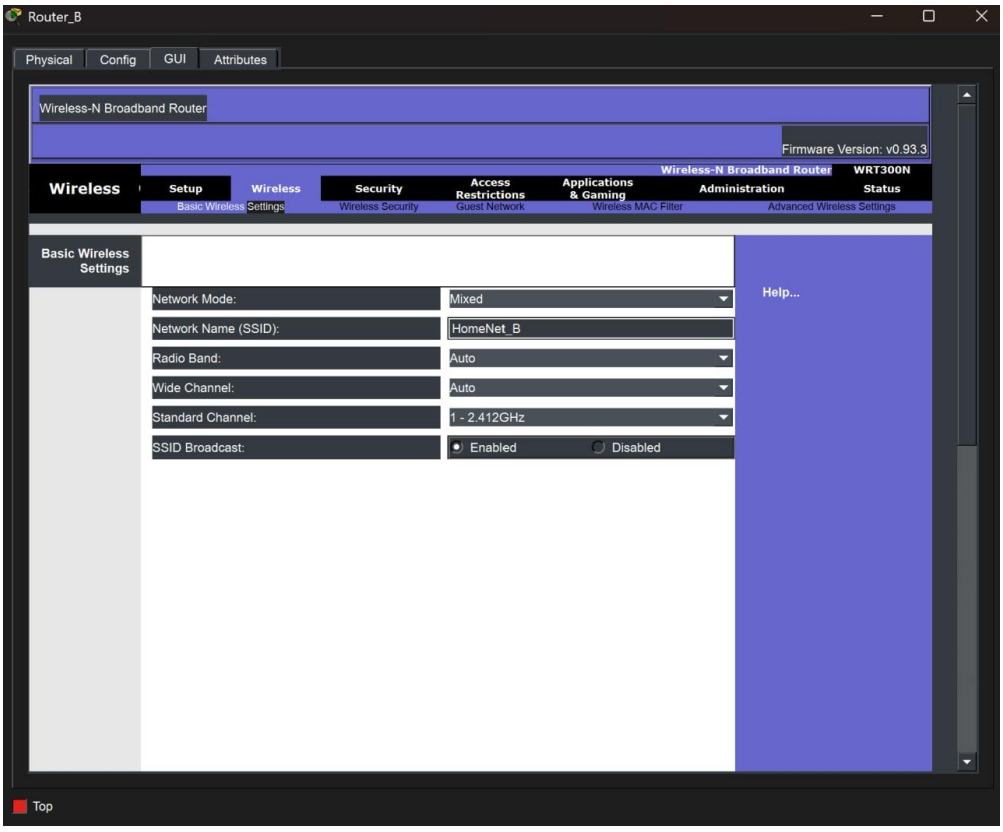


Description: This image illustrates the client-side wireless setup and successful workspace association. The laptop has been fitted with a wireless network module, and its configuration parameters match the router with the SSID HomeNet_A and the correct WEP key. As a result, the logical topology window shows all three client devices successfully connected to Router_A via active wireless links.



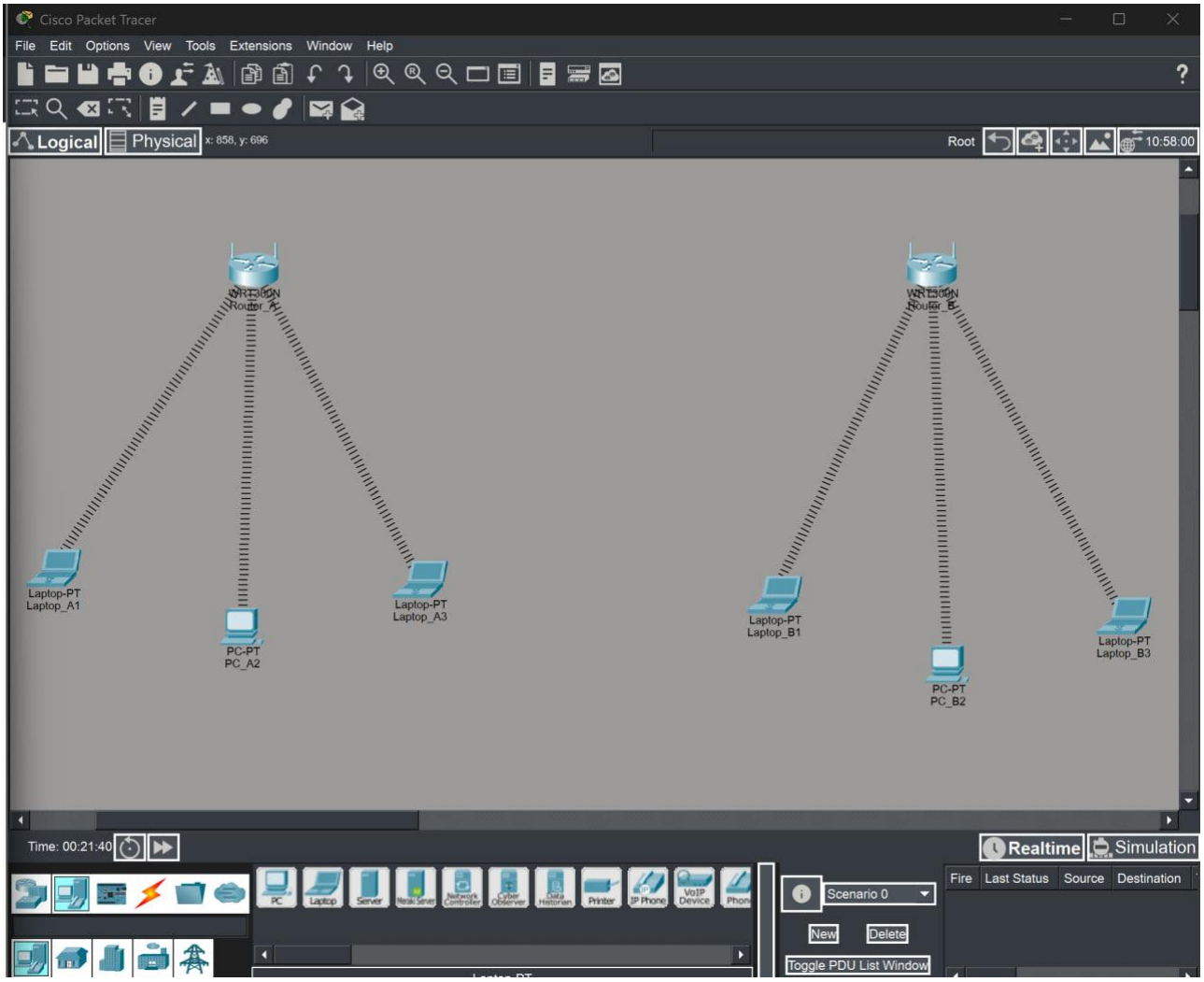
Description: This screenshot confirms connectivity within Network A. The ipconfig command verifies that an IP address was dynamically assigned by DHCP. A ping command to the default gateway (192.168.1.1) returns four successful replies with zero packet loss, confirming proper connectivity.



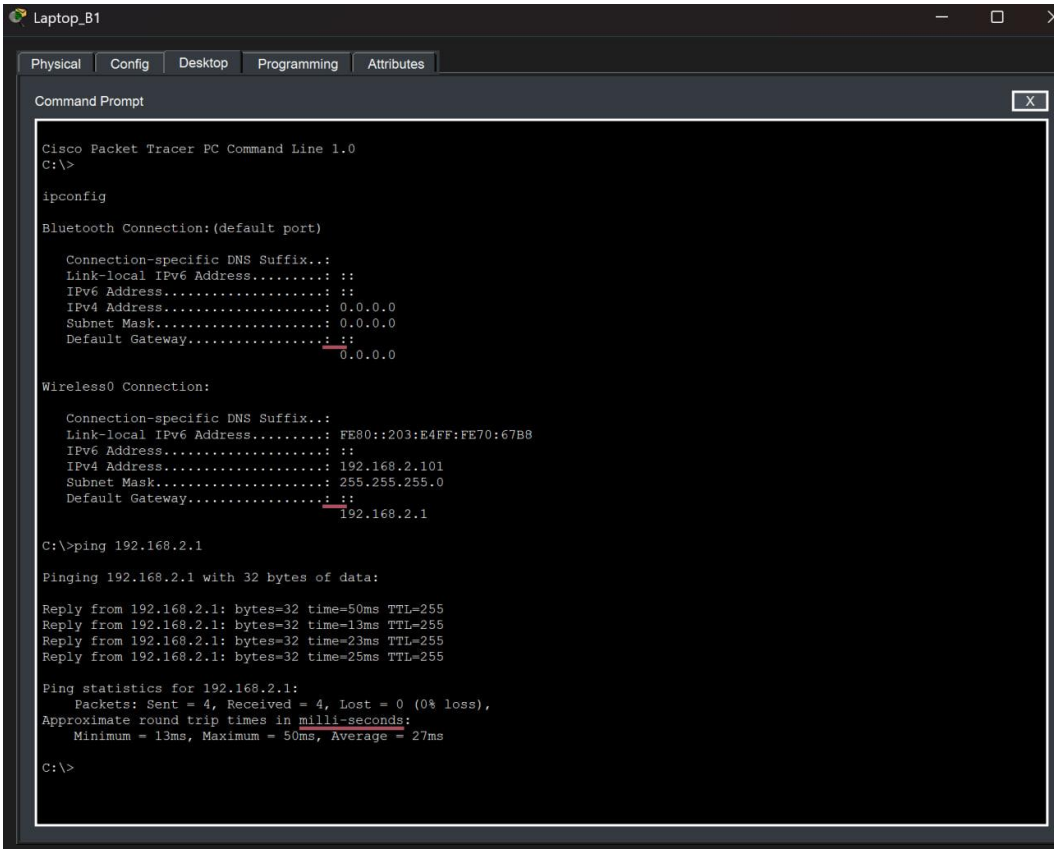


Description: This screenshot documents the setup and security settings for the second wireless router, Router_B. Under the basic configuration pages, the local gateway IP is adjusted to

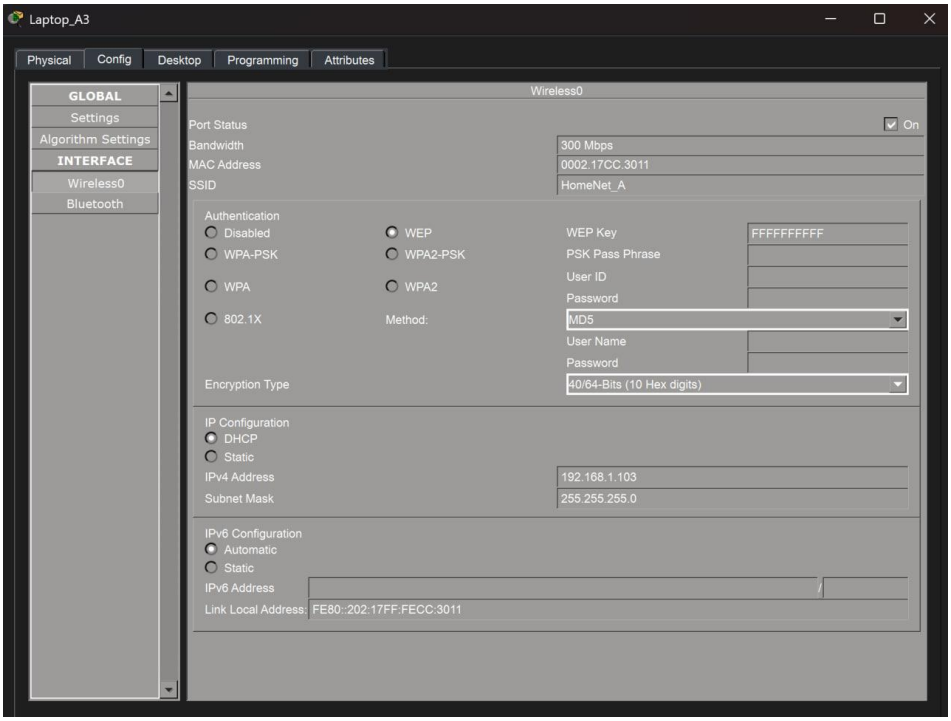
192.168.2.1 and the SSID name is set to HomeNet_B. Security is enabled by setting the encryption mode to WEP with a distinct 10-character hexadecimal password configured as Key 1: 1A2B3C4D5E.

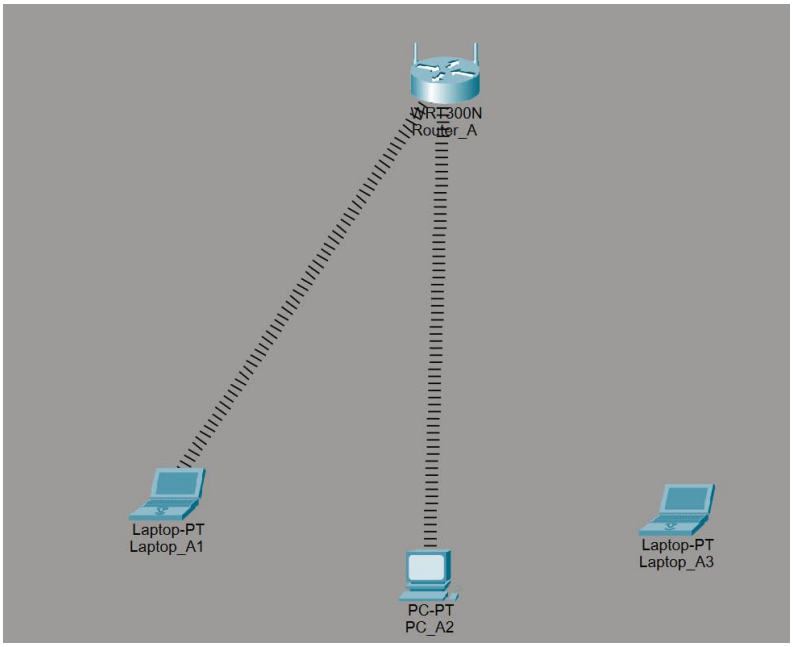


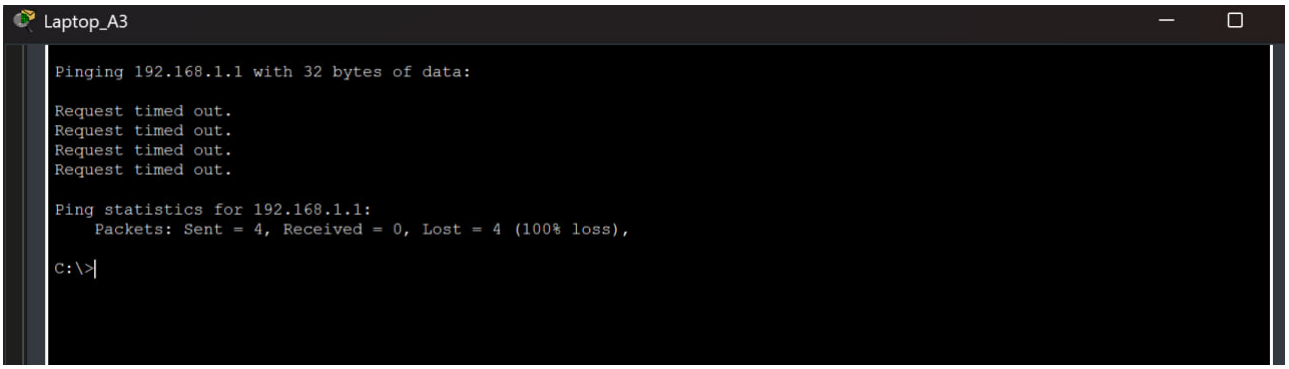
Description: This figure shows the final, completed network topology in the Cisco Packet Tracer logical workspace. It clearly shows both Network A (HomeNet_A) and Network B (HomeNet_B) functioning side-by-side as two completely independent environments. All client laptops and PCs have successfully verified their settings against their respective routers, establishing stable wireless links shown by the active dotted association lines.



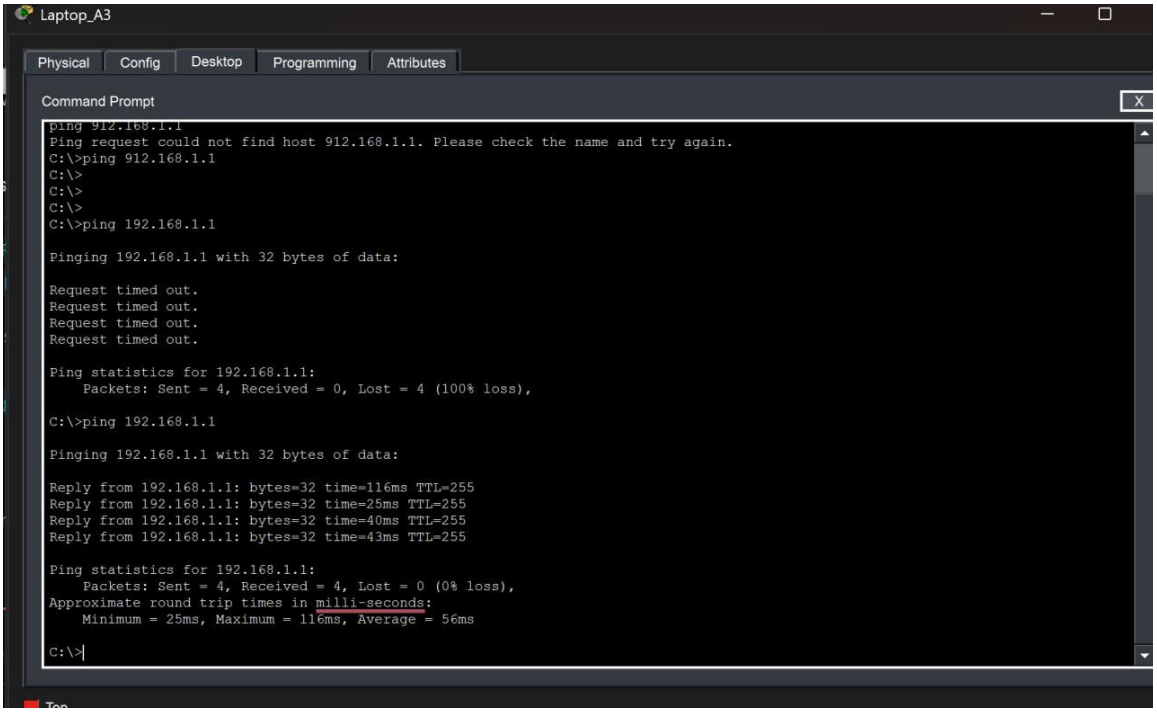
Description: This image verifies successful local network connection on the client side of Network B. The client machine correctly acquired an IP address inside the 192.168.2.x range via the router's active DHCP pool. Ping the gateway at 192.168.2.1 succeeds with replies and no dropped packets.







Description: These screenshots demonstrate the security enforcement of the WEP configuration via a negative test. When the WEP key on a client device from Network A is changed to an incorrect value (FFFFFFFFFF), the wireless link is dropped. The command prompt proves that network access is blocked, as the ping attempts fail completely and time out.



Restored connectivity.

Addressing Table:

Device/Setting	HomeNet_A	HomeNet_B
Gateway	192.168.1.1	192.168.2.1
Laptop1	192.168.1.100	192.168.2.101
PC2	192.168.1.102	192.168.2.102
Laptop3	192.168.1.103	192.168.2.103
Subnet Mask	255.255.255.0	255.255.255.0

Question 1: WEP encrypts traffic, yet we call it broken. Based on Session 2, what specifically makes WEP insecure even though the data is "encrypted"? (Mention the IV and keystream reuse.)

Even though WEP encrypts data, it is broken because it uses a very short 24-bit Initialization Vector (IV) sent in plain text. On busy networks, this short size forces the IVs to repeat, causing keystream reuse that allows attackers to easily calculate and crack the wireless password.

Question 2: In your negative test, a wrong key blocked the client. What does this demonstrate about WEP's notion of identity? Why can't WEP revoke a single user?

The negative test shows that WEP defines identity purely by possession of the correct password rather than identifying individual users or devices. Because every device on the network shares the exact same key, the router cannot block one single user without changing the password for everyone else at the same time.

Question 3: Both networks used a single shared key for all clients. Describe one real-world problem this causes in an organisation with many users.

If an employee leaves the organization or a laptop is stolen, the entire wireless network is immediately compromised because everyone shares the same key. The IT department must then manually update the password on hundreds of machines, creating massive administration issues and wasting business time.

Question 4: Name the modern scheme you would use instead of WEP today, and state one concrete improvement it provides (refer to WPA2 or WPA3 from the lecture).

Modern networks should use WPA2 or WPA3 security standards instead of outdated WEP encryption. These schemes improve security by giving every connected device its own unique, temporary dynamic session key to stop keystream reuse and prevent users from spying on each other.

This lab was completed by me in Cisco Packet Tracer.